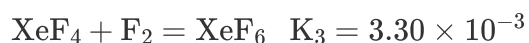


Question

Given the equilibrium constants (concentration equilibrium constants, units omitted) for the reaction of Xe with F₂ at 500°C:



At 500°C, Xe at an initial concentration of 0.100 mol/L and F₂ at an initial concentration of 0.200 mol/L react in a constant-volume container. What is the remaining concentration of F₂ when the reaction reaches equilibrium? (Retain two significant figures in the result)

A. All other options are incorrect

B. 0.070

C. 0.080

D. 0.090

E. 0.10

F. 0.11

G. 0.12

H. 0.13

I. 0.14

Answer

Correct Answer: **G**

Detailed Explanation

Let the equilibrium concentration of Xe be x , the concentration of F_2 be y , and the concentrations of XeF_2 , XeF_4 , XeF_6 be a , b , c respectively, then:

$$x + a + b + c = 0.1$$

$$y + a + 2b + 3c = 0.2$$

$$a = 29.8xy$$

CHECKPOINT

1 PTS

$$a = 29.8xy$$

$$b = 0.50ay = 29.8 \times 0.50xy^2$$

CHECKPOINT

1 PTS

$$b = 29.8 \times 0.50xy^2$$

$$c = 3.30 \times 10^{-3}by = 3.30 \times 10^{-3} \times 29.8 \times 0.50xy^3$$

CHECKPOINT

1 PTS

$$c = 3.30 \times 10^{-3} \times 29.8 \times 0.50xy^3$$

Substituting the last three equations into the first two equations, we have:

$$x(1 + 29.8y + 14.9y^2 + 0.0492y^3) = 0.1$$

$$y(1 + 29.8x + 29.8xy + 0.1476xy^2) = 0.2$$

The first equation is transformed into $x = \frac{0.1}{1+29.8y+14.9y^2+0.0492y^3} = \frac{\frac{0.2}{y}-1}{29.8+29.8y+0.1476y^2}$

CHECKPOINT

1 PTS

$$x = \frac{\frac{0.2}{y}-1}{29.8+29.8y+0.1476y^2}$$

Substituting x into the second equation, we obtain $y = 0.117$

CHECKPOINT

1 PTS

$$y = 0.117$$